FitnessBalance — database structure overview

Logical Firestore model • key attributes shown • private documents inherit ownership from users/{uid}

users
user-owned document

uid [ID]
fullName
email

foods
shared catalogue

foodKey [ID]
food_name
source_database

Nutrition map
embedded map / record

energy_kcal
protein_g

foodLogs
user-owned document

date [ID]
date
foodItems

Food entry snapshot
embedded map / record

entryId
foodId

dailyData
user-owned document

date [ID]
caloriesConsumed
caloriesBurned

exercises
shared catalogue

exerciseId [ID]
name
category

workoutPaths
shared catalogue

pathId [ID]
name
active

lessons
shared catalogue

lessonId [ID]
name
sequence

workoutExercises
shared catalogue

itemId [ID]
exerciseId
sequence

workoutSessions
user-owned document

sessionId [ID]
pathId
lessonId

workoutRecords
user-owned document

sessionId [ID]
pathId
lessonId

exerciseLogs
user-owned document

date [ID]
date
exercises

Activity entry
embedded map / record

entryId
name

savedMeals
user-owned document

mealId [ID]
name
items

recipes
user-owned document

recipeId [ID]
name
items

personalServings
user-owned document

servingId [ID]
foodKey
name

barcodeFoods
user-owned document

barcode [ID]
food_name
nutrition_basis

foodNutrition
user-owned document

foodId [ID]
originalFoodId
nutrition

weightLogs
user-owned document

date [ID]
day
weightKg

weightHistory
user-owned document

state [ID]
entries
updatedAt

pathProgress
user-owned document

pathId [ID]
completionCounts
bestPercent

exerciseWeeks
user-owned document

week [ID]
completed
target

quests
shared catalogue

questId [ID]
name
target

questProgress
user-owned document

date_questId [ID]
day
questId

achievements
shared catalogue

achievementId [ID]
name
description

unlockedAchievements
user-owned document

achievementId [ID]
recordId
unlockedAt

savedMealLogs
user-owned document

operationId [ID]
date
mealId

wearableWorkoutLinks
user-owned document

entryId [ID]
date
entryId

wearableWorkoutRewards
user-owned document

entryId [ID]
entryId
day

wearableRewardDays
user-owned document

date [ID]
day
week

ID = document path key; dashed = logical reference; dotted = derived/index data; green solid = embedded structure.
This overview lists all 31 depicted structures, including three embedded structures. Relationship lines and cardinalities are shown on the following focused ERD pages. Firestore has no SQL foreign-key constraints. Source references can survive deletion.

FitnessBalance — food and nutrition ERD

Logical Firestore model • key attributes shown • private documents inherit ownership from users/{uid}

originalFoodId: 1 — 0..*

foods
shared catalogue
- foodKey [ID]
- food_name
- source_database
- nutrition_basis
- nutrition

embeds
1 — 1

Nutrition map
embedded map / record
- energy_kcal
- protein_g
- carbohydrate_g
- fat_g

foodNutrition
user-owned document
- foodId [ID]
- originalFoodId
- nutrition
- nutrition_basis
- source_kind

dailyData
user-owned document
- date [ID]
- caloriesConsumed
- caloriesBurned
- nutrientTotals
- structuredExerciseContributions

foodId
0..1 — 0..*

barcodeFoods
user-owned document
- barcode [ID]
- food_name
- nutrition_basis
- nutrition
- label_nutrition

product key
0..1 — 0..*

personalServings
user-owned document
- servingId [ID]
- foodKey
- name
- amount
- unit

savedMeals
user-owned document
- mealId [ID]
- name
- items
- revision

mealId
0..1 — 0..*

foodLogs
user-owned document
- date [ID]
- date
- foodItems
- totalCalories
- nutrientTotals

embeds
1 — 0..*

Food entry snapshot
embedded map / record
- entryId
- foodId
- name
- meal

recipeId
0..* — 0..1

recipes
user-owned document
- recipeId [ID]
- name
- items
- yieldServings
- revision

source
0..1 — 0..*

savedMealLogs
user-owned document
- operationId [ID]
- date
- mealId
- entryCount

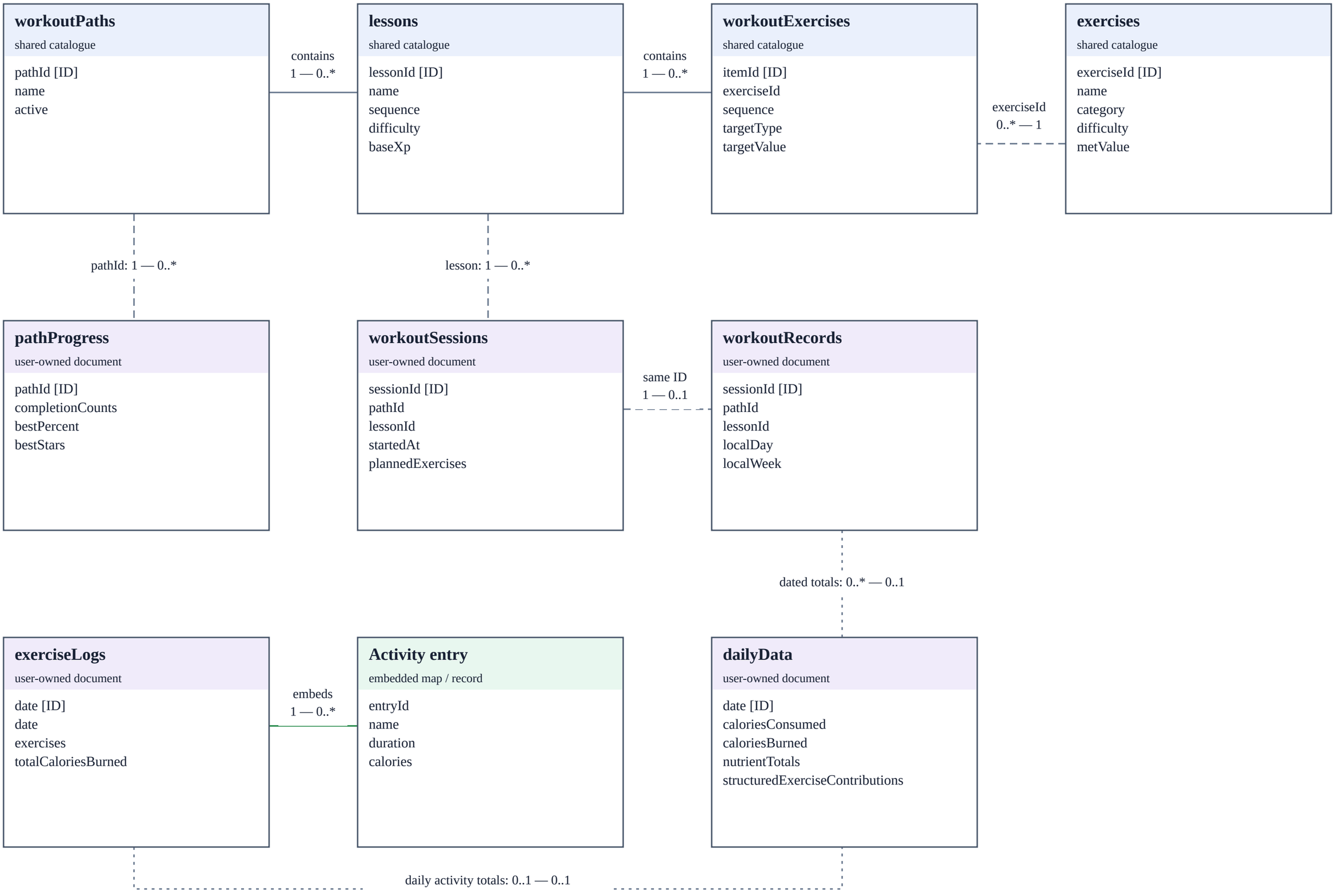
savedMealId: 0..* — 0..1

derived totals by user/date: 0..1 — 0..1

ID = document path key; dashed = logical reference; dotted = derived/index data; green solid = embedded structure.
Private barcode foods and catalogue foods are alternative sources for personal portions. Recipe/meal reuse stores historical snapshots. Additional source references are omitted here to avoid crossing lines; the data dictionary retains their fields.

FitnessBalance — exercise and workout ERD

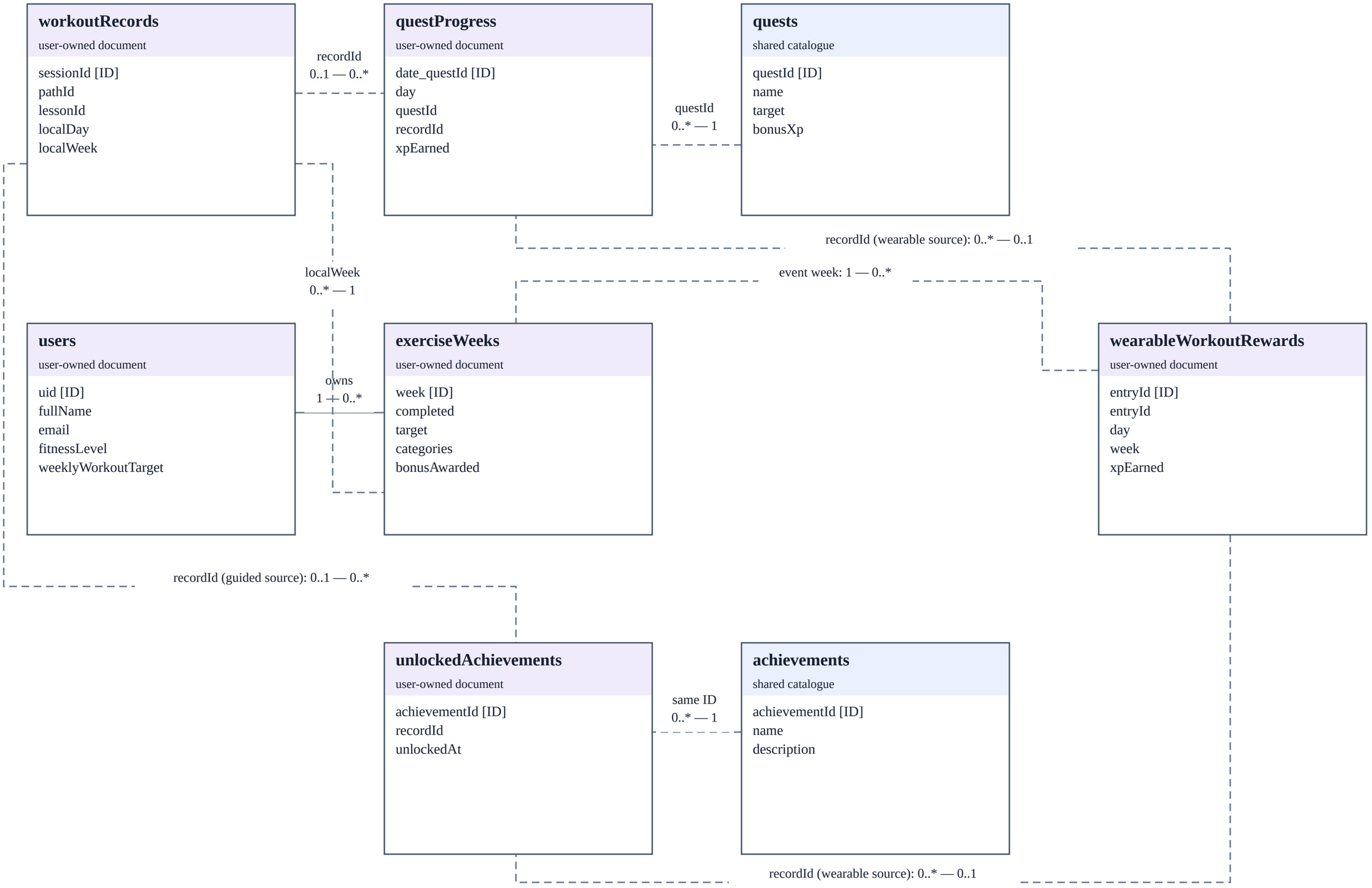
Logical Firestore model • key attributes shown • private documents inherit ownership from users/{uid}



ID = document path key; dashed = logical reference; dotted = derived/index data; green solid = embedded structure.
A lesson reference includes pathId and lessonId. A started session may have no completed record. Guided results and manual/wearable activity entries use separate source documents but contribute to daily summaries.

FitnessBalance — quests, achievements and weekly progress ERD

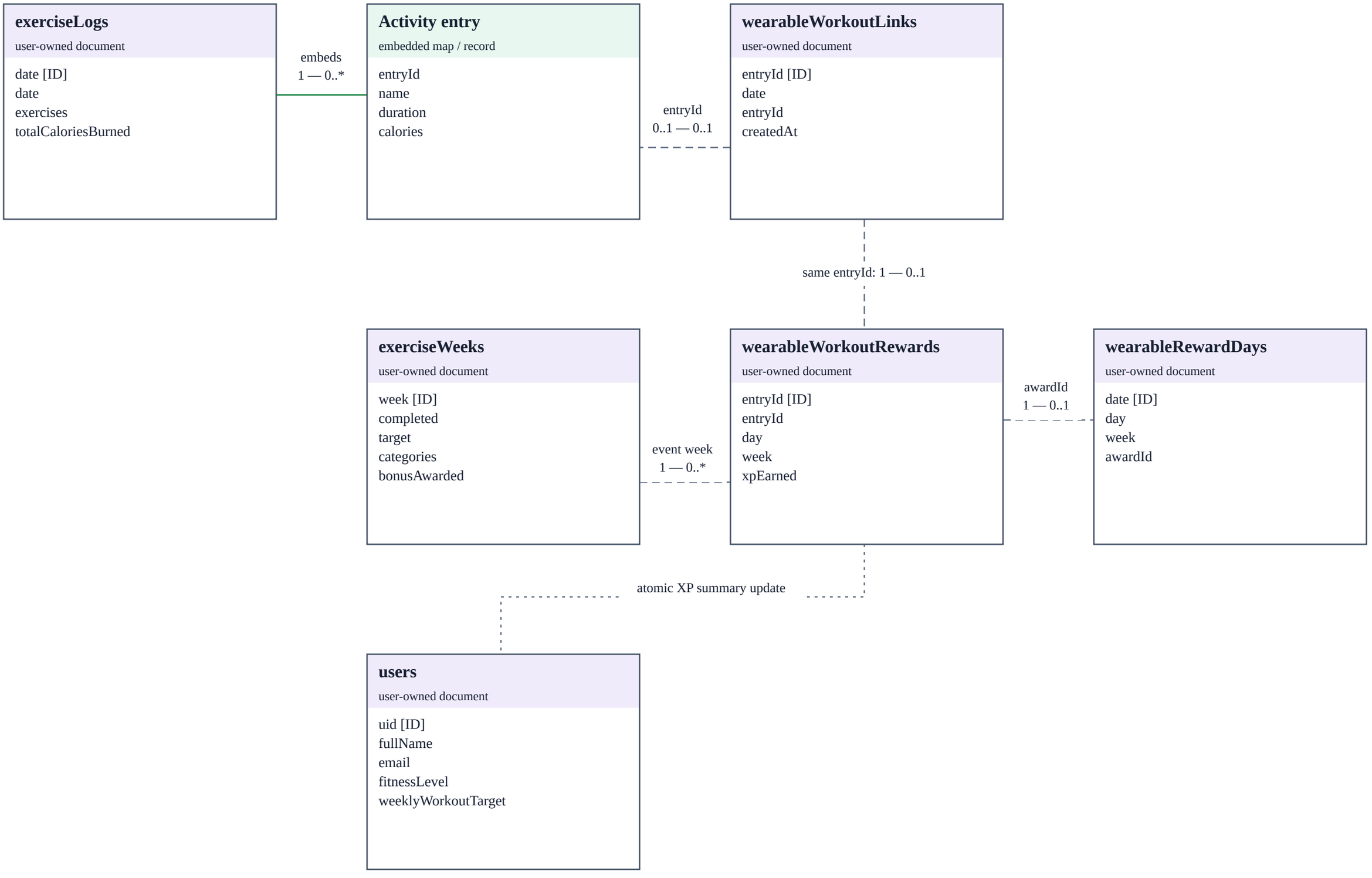
Logical Firestore model • key attributes shown • private documents inherit ownership from users/{uid}



ID = document path key; dashed = logical reference; dotted = derived/index data; green solid = embedded structure.
recordId refers to a guided workout OR a wearable award, not both as mandatory parents. Only one user-ownership edge is drawn; all private progress remains nested under users/{uid}. Immutable award evidence prevents repeated rewards.

FitnessBalance — wearable confirmation and reward ERD

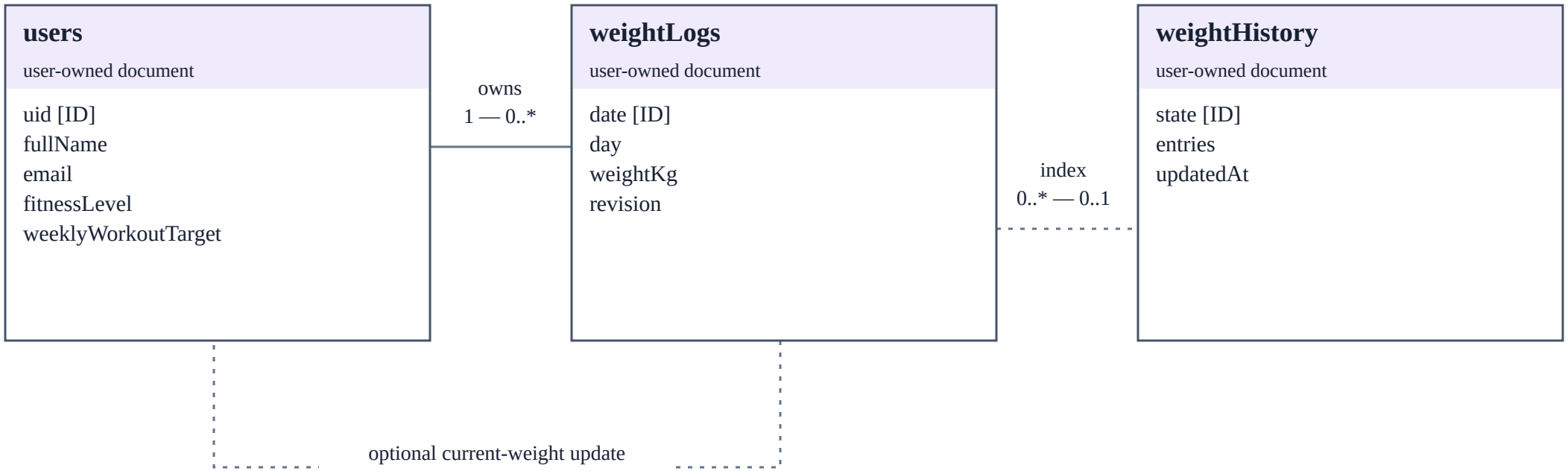
Logical Firestore model • key attributes shown • private documents inherit ownership from users/{uid}



ID = document path key; dashed = logical reference; dotted = derived/index data; green solid = embedded structure.
Only wearable entries have confirmation receipts; manual entries do not. An eligible confirmed session may have one reward ledger. A date-credit receipt identifies its first award and survives activity deletion. Health Connect caches stay on-device until explicit confirmation.

FitnessBalance — weight history ERD

Logical Firestore model • key attributes shown • private documents inherit ownership from users/{uid}



ID = document path key; dashed = logical reference; dotted = derived/index data; green solid = embedded structure.
Each dated weight document has one transaction-maintained entry in the private weightHistory/state index. There is at most one state index per user.
Updating profile weight is optional; it does not replace historical measurements.